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18 May 2026

Prof. Dr. Andre Franke
Institut für Klinische Molekularbiologie (IKMB)
Universitätsklinikum Schleswig-Holstein, Campus Kiel
Ref. 28608

Re: PhD Position — Computational Immunology / IBD Genomics (Ref. 28608)

Dear Prof. Franke,

I am applying for the PhD position in computational immunology at IKMB (Ref. 28608). The position covers immune repertoire analysis, single-cell sequencing, and the microbiome–host genetics interface in intestinal inflammation — precisely the application domain for a computational immunopeptidology toolkit I have developed and deployed over the past four years. The specific analytical questions the position describes — which epitopes drive mucosal T cell responses, how immune repertoires shift across IBD patient cohorts, how host genetic variation modulates inflammatory microbiome interactions — are computationally tractable with the spectral methods and Bayesian inference infrastructure I have built.

Computational immunopeptidology for IBD T cell biology. The Syndrome framework applies 3-channel discrete Fourier transforms (hydropathy, volume, charge) to compute spectral complementarity ρ between any two molecular sequences in $\mathcal{O}(cL \log L)$. For IBD, the three directly applicable tools are:

- **MHC binding predictor:** $\rho \geq 0.72$ for binding prediction; AUC = 0.81 vs. NetMHCpan = 0.68 on 2,847 IEDB peptides ($r = 0.73$, $p < 10^{-12}$). Applicable to patient-specific HLA scanning: given a patient's HLA haplotype and the microbial peptide landscape from their metagenomics data, predict which microbial epitopes will be presented and drive mucosal T cell activation.
- **Neoantigen / cross-reactive epitope identification:** dual threshold $\Delta\rho_{\text{self}} > 0.15$ (sufficient divergence from self) and $\rho_{\text{MHC}} > 0.72$ (sufficient binding affinity). In IBD, this identifies microbial epitopes with high cross-reactivity risk to intestinal self-antigens — the computational screen for molecular mimicry candidates.
- **Escape predictor:** systematic single amino acid substitution scan scoring $E(v) = -\max \rho(v, \text{self}) + w \cdot \rho(v, \text{target})$. For immune repertoire analysis, this maps how observed TCR sequences relate to the epitope landscape: which receptor variants maintain target recognition while minimising autoreactive cross-reactivity.

These tools are live at syndrome-seven.vercel.app and require no local installation, making them directly usable by wet-lab collaborators without computational infrastructure.

IBD genomics and microbiome–host genetics. The Gospel framework implements

large-scale genomics analysis at 10^6 variants per second: somatic and germline variant calling, RNA-seq quantification and differential expression, Bayesian network inference for multi-omics integration, and Gaussian mixture models for population structure. Validated against 1000 Genomes Phase 3, gnomAD v3.1.2, and UK Biobank — the same cohort scales as IBD GWAS studies. The Bayesian network module performs multi-omics integration directly: host genotype variation as upstream nodes conditioning metagenomics and transcriptomics distributions downstream, with posterior inference across the joint graph. This is the computational structure for microbiome–host genetics analysis: identifying which host genetic variants (IBD GWAS loci) modulate microbial community composition and mucosal inflammatory gene expression simultaneously, rather than correlating the two separately.

Disease state trajectory and therapeutic target selection. The Syndrome framework’s disease state layer represents patient state as a vector $\mathbf{D} \in [0, 1]^8$ with dimensions including immune activation (D_{Imm}), mucosal barrier integrity (D_{Bar}), ATP synthesis coherence (D_{ATP}), and circadian regulatory coherence (D_{Circ}). Trajectory inference uses a Bayesian $P(R | Q)$ posterior gating structure, giving a principled model of disease progression rather than a cross-sectional score. For minimum-intervention target identification — which single pathway perturbation most efficiently restores the disease state vector toward remission — the framework uses a sparse ℓ_1 linear program, validated at $\text{AUC} = 1.00$ across 25/25 validation experiments. In the IBD context, this provides a computational framework for personalised therapeutic target prioritisation from single-cell multi-omics data: patient state characterisation from scRNA-seq and TCR-seq jointly, followed by LP-based identification of the minimal intervention set.

Single-cell methods and repertoire analysis. My primary single-cell experience is through the Bayesian inference layer: variational Bayes posterior approximation at $\mathcal{O}(10^6)$ latent variable scale (Gospel), probabilistic graphical models for multi-omics integration, and Gaussian processes for trajectory inference. Scanpy and Seurat are familiar from MSc training and independent study; the statistical models underlying UMAP-based clustering and RNA velocity are directly related to the Bayesian dimensionality reduction infrastructure I have built. TCR/BCR clonotype analysis — the combinatorial structure of V(D)J recombination, clonal expansion statistics, and cross-patient repertoire comparison — is a sequence analysis problem where the spectral complementarity methods in Syndrome apply directly: ρ as a distance measure between TCR sequences, enabling clustering of functionally similar clonotypes without requiring MSA. The PhD provides the experimental immunology context (patient samples, immune phenotyping, metagenomics protocols) that gives these computational methods their biological grounding.

Background. I hold an MSc in Computational Biology (Universität Tübingen, 2018–2019) and conducted doctoral research in Computational Lipidomics at TUM/Bayerisches Forschungsinstitut (2019–2021), working on ML for large-scale MS omics data in clinical cohorts. I am legally resident in Germany with full work authorisation. English is native; German is conversational.

Yours sincerely,

Kundai Farai Sachikonye